Pre-Analysis Plan for #ManyDaughters

RT:

Introduction

This Pre-Analysis Plan (PAP) is for the #ManyDaughters research project. The goal of this PAP is to outline the data from the German Socio-Economic Panel (SOEP) which will be used to test four (4) hypotheses surround the effect of having daughters on four different social and political outcomes.

At the writing of this PAP, we do not have access to the SOEP survey data. Various instructional resources such as `https://paneldata.org/` and `https://companion.soep.de/` were used to gain a better understanding of the underlying data and identify the variables of interest for testing the four hypotheses.

Datasets

We will have access to eight (8) datasets within the SOEP-Core survey results: *pl, hl, biol, biobirth, bioparen, pgen, hgen, and ppathl*. The data used to test the hypotheses will only make use of one (1) of these: *pl*. As previously mentioned, we do not have access to the underlying data when writing the PAP. When variable answers were only listed in German on `https://paneldata.org/`, Google Translate was used to translate the answers into English.

Outcome Variables

Hypotheses 1 and 4 (H1 and H4) will have "Gender Equality Attitudes" as the outcome variable: y_{it}^{GEA}. In the *pl* dataset these will be created via: `plj0621`, `plj0622`, `plj0623`, `plj0624`, and `plj0625`. These as the following five questions, all answered on a 0–7 scale, with 0 being "Strongly Disagree" and 7 being "Strongly Agree":

- `plj0621`: "Having a job is the best way for a woman to be independent." [Strongly Agree means *more* gender equity]
- `plj0622`: "Even a married woman should have a paid job so that she can be financially independent." [Strongly Agree means *more* gender equity]
- `plj0623`: "If a woman earns more money than her partner, this inevitably leads to problems." [Strongly Agree means *less* gender equity]
- `plj0624`: "For parents, vocational training or higher education for their sons should be more important than vocational training or higher education for their daughters." [Strongly Agree means *less* gender equity]
- `plj0625`: "At home, the husband should have the final say." [Strongly Agree means *less* gender equity]

As `plj0623`, `plj0624`, and `plj0625` are reverse-coded, their 0–7 scale will be "flipped", and then the average of all five variables will be taken to produce a measure General Equality Attitudes. Any 'negative' values will be dropped, as they imply non-answers or missing answers. Although choosing the majority of variables to be re-coded, this will allow for the more 'natural' interpretation of having more daughters *increasing* support for gender equality.

Hypothesis 2 (H2) will have "Political Attitude" (y_{it}^{PA}) as the outcome variable. In the *pl* dataset, `plh0004` will be directly used as the outcome variable. The question asks, "In politics people often talk about 'left' and 'right' when it comes to characterize different political attitudes. If you think about your own political

views: Where would you place yours?" where 0 is 'far-left' and 10 is 'far-right'. Any 'negative' values will be dropped, as they imply non-answers or missing answers. The data will be reversed prior to analysis as once again this will allow for the more 'natural' interpretation of having more daughters *increasing* left-leaning political beliefs.

Hypothesis 3 (H3) will have 'Household Gender Equity" (y_{it}^{GEH}) as the outcome variable. In the *pl* dataset, three variables will be used: `pli0019_h`, `pli0022_h`, and `pli0044_h`. These all ask the same question: "What is a typical day like for you? How many hours do you spend on the following activities on a typical [Saturday (`0019`)), Sunday (`0022`), Weekday (`0044`)]? One of the actives listed is childcare. The weekday value will be multiplied by five, and then added to the value from Saturday and Sunday to produce the total hours spent on childcare in a week. The '_h' in the variable names indicates that the values have been harmonized over the course of the survey. If all three values are not available, only the weekday value will be used.

Identifying Variables

As we are interested in the effect of having daughters, we'll use a variable to identify the presence of having daughters. And we will also need to identify the gender of the respondent.

- "Daughters" will be measured via `pld0025` which asks "And now about your close and extended family: Which and how many of the following relatives do you have? Please also state where they live." A value of -2 means the respondant does not have any daughters and will be re-coded to 0.
- "Gender" will be measured by `pla0047_v2`, which asks "What gender do you identify as?". Only respondents who answer "Male" or "Female" will be kept and then re-coded as a 0-1 variable.

Control Variables

Given that attitudes toward gender can be influenced by other factors, some control variables will be included. The four (4) controls will be: "Income", "Education", "Marital Status", and "Year"

- "Income" ($Income_{it}$) will be `plc0013_h` which asks, "What did you earn from your work last month?" and has been harmonized to account for the introduction of the Euro.
- "Education" ($Education_{it}$) will be measured by `plg0076` which asks "Did you graduate with a degree upon finishing your studies?" and is therefore a 0-1 variable.
- "Martial Status" (δ_{it}^{M}) will be measured by `plj0626` which asks "What is your marital status?" There are seven potential answers and some will be grouped together: Married and Registered Partnership will be coded as the same. Divorced and Annulled will be coded as the same. Lastly, Widowed and Deceased Registered partner will be coded as the same. This variable will then be used as a fixed-effect for each of the four types.
- "Year" (δ_{t}^{T}) will be from `syear` which is the year the survey was conducted. The variable will be used as a fixed-effect.

Inclusion/Exclusion Criteria

When given access to the dataset, the only observations which will be dropped when testing a specific hypothesis are observations which are missing values for the variables of interest, with the only except being someone who did not answer the Male/Female binary for gender. As the data will cover 2000–2020, there is no need to drop observations from pre-unification of Germany. Some hypotheses might use subsets of the data, and they will be detailed in the next section.

Regression Specification and Hypotheses

Four hypotheses will be tested, the hypothesis, its specification are:

1. **Hypothesis 1:** Having daughters leads to stronger support for **G**ender **E**quality (as measured by **A**ttitudes) will be estimated by

$$y_{it}^{GEA} = \alpha + \beta_1 Daughters_{it} + \beta_2 Income_{it} + \beta_3 Education_{it} + \delta_{it}^{M} + \delta_t^{T} + \epsilon_{it}$$

   where $Daughters_{it}$ is a zero-one variable that takes the value of one if the respondent (i) has at least one daughter in year t. The outcome variable y_{it}^{GEA} is a construction from five questions about the role of women in society. **This means β_1 is the coefficient of interest, with our hypothesis implying $\beta_1 > 0$.** *Income* and *Education* are controls for the respondents income and education. δ_{it}^{M} and δ_t^{T} are fixed-effects for marital status and year. ϵ_{it} is the error term and will be calculated with robust SEs. Although the dependent variable is a Likert-scale, the estimation will still use OLS, and instead only focus on the sign of β_1.

2. **Hypothesis 2:** Having daughters leads to more left-leaning political preferences will be estimated by

$$y_{it}^{PA} = \alpha + \beta_1 Daughters_{it} + \beta_2 Income_{it} + \beta_3 Education_{it} + \delta_{it}^{M} + \delta_t^{T} + \epsilon_{it}$$

   where $Daughters_{it}$ is a zero-one variable that takes the value of one if the respondent (i) has at least one daughter in year t. The outcome variable y_{it}^{PA} where respondants rank them self on a right-left scale. **This means β_1 is the coefficient of interest, with our hypothesis implying $\beta_1 > 0$.** *Income* and *Education* are controls for the respondents income and education. δ_{it}^{M} and δ_t^{T} are fixed-effects for marital status and year. ϵ_{it} is the error term and will be calculated with robust SEs. Although the dependent variable is a Likert-scale, the estimation will still use OLS, and instead only focus on the sign of β_1.

3. **Hypothesis 3:** Having daughters leads to more **G**ender **E**quality in the **H**ousehold (as measured by behaviors or outcomes), which will be estimated by

$$y_{it}^{GEH} = \alpha + \beta_1 Daughters_{it} + \beta_2 Income_{it} + \beta_3 Education_{it} + \delta_t^{T} + \epsilon_{it}$$

   however, given this hypothesis is measured at the household level, the sample will be restricted to married men. The idea being in a household with more gender equity, the husband will do more childcare—y_{it}^{GEH} measures the total hours a week the respondent puts towards providing childcare. **This means β_1 is the coefficient of interest, with our hypothesis implying $\beta_1 > 0$.** *Income* and *Education* are controls for the respondents income and education. δ_t^{T} are fixed-effects for year. ϵ_{it} is the error term and will be calculated with robust SEs. Given the count nature of hours spent on childcare, this will be estimated with a Poisson.

4. **Hypothesis 4:** Having daughters leads to a larger increase in the support of gender equality for men than for women (with **G**ender **E**quality measured by **A**ttitudes) will be estimated by

$$y_{it}^{GEA} = \alpha + \beta_1 Daughters_{it} + \beta_2 Male_i + \beta_3 (Daughters_{it} \times Male_i) + \beta_4 Income_{it} + \beta_5 Education_{it} + \delta_{it}^{M} + \delta_t^{T} + \epsilon_{it}$$

   where $Daughters_{it}$ is a zero-one variable that takes the value of one if the respondent (i) has at least one daughter in year t and $Male_i$ is a zero-one variable that takes the value of one if the respondent

is a male. The outcome variable y_{it}^{GEA} is a construction from five questions about the role of women in society. **This means β_3 is the coefficient of interest, with our hypothesis implying $\beta_3 > 0$.** *Income* and *Education* are controls for the respondents income and education. δ_{it}^{M} and δ_{t}^{T} are fixed-effects for marital status and year. ϵ_{it} is the error term and will be calculated with robust SEs. Although the dependent variable is a Likert-scale, the estimation will still use OLS, and instead only focus on the sign of β_1.